

Users' Perception Analysis of Cyberattack Consequences Using NLP

Yasaman Ebrahimi Bao Hong Gia Nguyen
CS 273 Project 2.2

I. INTRODUCTION

Project Overview: This project aimed at comparing human judgments of the consequences of cyber-attacks and similarity values that can be obtained from the NLP methods. The human data was obtained from a card-sorting process and the NLP data was based on the text of the 50 consequence descriptions. We were interested to see if the consequences that were assigned to similar groups would be similar when compared with BOW, TF-IDF and WordNet.

The project consisted of three main experiments. We first constructed the human based similarity matrix by Spearman correlation. Second, we generated 12 BOW matrices and 12 TF-IDF matrices with the necessary preprocessing settings as well as cosine similarity. Third, four WordNet matrices were produced following the desired procedure as used in GLAO. All 28 NLP matrices were then correlated with the human matrix using a Pearson correlation.

II. DATA DESCRIPTION AND LIBRARIES

The data was taken from `all_data.xlsx`. The worksheet analyzed has a total of 226 rows and 53 columns. The first two columns are the participant and group information while the remaining 50 columns are cyberattack consequence descriptions. Each row of consequences is also a 226-value binary vector that was derived from the card-sorting data. If the result is 1, then the consequence was present in the group represented by that row; if the result is 0, it is not present. This enabled the column vectors of the human experiment to be used as the text documents for the NLP experiments and the column names to be used as the vector of the human experiment.

The notebook was written in Python programming language. The most important libraries that were used are NumPy for data, pandas for data, SciPy for Spearman and Pearson correlation, scikit-learn for CountVectorizer, TfidfVectorizer, and cosine similarity, NLTK for tokenization, POS tagging, stopwords, stemming, lemmatization, and WordNet, and Matplotlib for heat maps. We also utilized tools from `itertools`, `functools`, and `IPython` to structure repeated configurations and show the outcomes.

III. EXPERIMENT 1: HUMAN-BASED SIMILARITY MATRIX

In the first experiment, the 226-values card-sorting vectors provided with each cyberattack consequence were used.

Spearman's rank correlation was used to compare each pair of consequence vectors. The calculation can be written as

$$\rho_s(x, y) = \text{corr}(R(x), R(y)), \quad (1)$$

where $R(x)$ and $R(y)$ are the ranked versions of the two vectors. For the binary vectors, we used the implementation from SciPy since many values are tied.

The resulting matrix is a 50×50 matrix that is symmetric and has 1 on the diagonal. As shown in Fig. 1(a), there are positive and negative relationships in the human matrix. It also includes several visible blocks of consequences that were handled in a similar fashion by the participants.

IV. EXPERIMENT 2: BOW AND TF-IDF

A. Preprocessing Configurations

In the case of BOW and TF-IDF, all consequence descriptions were lowercased, tokenized and reduced to alphabetic tokens. All variations of stopwords retained or removed, stemming on or off, and the use of unigrams, bigrams, or trigrams were then tested. The result was $2 \times 2 \times 3 = 12$ configurations for BOW and 12 configurations for TF-IDF.

BOW is a representation of each consequence by the number of the selected words or n -grams. TF-IDF has the same general features, but assigns less weight to features found in many documents. Each pair of consequence vectors was compared via cosine similarity after vectorization,

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}. \quad (2)$$

These were thus each followed by their own 50×50 similarity matrix.

V. EXPERIMENT 3: WORDNET SIMILARITY

The WordNet experiment went through the necessary seven steps. The aim was to compare sentences based on the meaning rather than solely on a literal match of the words. The following steps in the notebook were:

- 1) Tokenize each sentence and assign Penn Treebank POS tags with `word_tokenize()` and `pos_tag()`.
- 2) Convert the Penn tags to WordNet tags `n`, `v`, `a`, or `r`. Tag `s` was treated as `a`, and unsupported tags were removed.
- 3) Retrieve the first WordNet synset for each supported token using `wn.synsets()`.
- 4) Compare synset pairs using WordNet `path_similarity()`.

- 5) For each source synset, keep its best target match, sum the valid scores, and divide by the maximum number of synsets in the two sentences.
- 6) Compute the score in both directions and average the two values so the result is symmetric.
- 7) Repeat the process for every pair of the 50 consequence descriptions to build the similarity matrix.

If A and B are the two synset lists, the directional score is

$$S(A, B) = \frac{\sum_{a \in A} \max_{b \in B} \text{path}(a, b)}{\max(|A|, |B|)}, \quad (3)$$

and the final symmetric score is

$$S_{\text{sym}}(A, B) = \frac{S(A, B) + S(B, A)}{2}. \quad (4)$$

The WordNet configurations tested were: stopwords with or without and lemmatization with or without. This yielded four WordNet similarity matrices.

VI. MATRIX COMPARISON

The upper triangle of each NLP matrix, excluding the diagonal, was compared with the human matrix. This equates to 1,225 different pairs of consequences. The same pair positions were used for all matrices and Pearson correlation was calculated,

$$r = \frac{\sum_i (h_i - \bar{h})(n_i - \bar{n})}{\sqrt{\sum_i (h_i - \bar{h})^2} \sqrt{\sum_i (n_i - \bar{n})^2}}, \quad (5)$$

where h is the human similarity values and n is the NLP similarity values. As the absolute value of the numbers in the NLP matrix increases, the closer it is to the human pairwise pattern.

VII. RESULTS

Figure 1 presents the human matrix and the optimum matrices in each NLP technique. In all four panels the consequence order is identical, which helps to visually compare the main blocks.

A. BOW Results

All 12 BOW configurations are listed in Table I. The best correlation was **0.599** with stopword removal and no stemming and unigrams. The unigram results were slightly affected by the change of stemming. The removal of stopwords did a bit better on the unigram, but it made the bigram and trigram results much lower. The longer word sequences were not as helpful as the individual content words in this data set.

B. TF-IDF Results

All 12 TF-IDF configurations are presented in Table II. When using the same method, but not stemming the words, the highest TF-IDF correlation was **0.555**. It was much like BOW overall. The unigrams, with stopword removal, were the best performers, whereas the bigrams and trigrams were worse. BOW outperformed TF-IDF in the best configuration, indicating that some of the common words found in the consequence descriptions remained relevant for the human groupings.

TABLE I
BOW CORRELATION WITH THE HUMAN MATRIX

Stopwords	Stemming	n -gram	Pearson r
Kept	No	Unigram	0.522
Kept	No	Bigram	0.529
Kept	No	Trigram	0.415
Kept	Yes	Unigram	0.521
Kept	Yes	Bigram	0.526
Kept	Yes	Trigram	0.413
Removed	No	Unigram	0.599
Removed	No	Bigram	0.382
Removed	No	Trigram	0.172
Removed	Yes	Unigram	0.595
Removed	Yes	Bigram	0.378
Removed	Yes	Trigram	0.172

TABLE II
TF-IDF CORRELATION WITH THE HUMAN MATRIX

Stopwords	Stemming	n -gram	Pearson r
Kept	No	Unigram	0.497
Kept	No	Bigram	0.487
Kept	No	Trigram	0.396
Kept	Yes	Unigram	0.489
Kept	Yes	Bigram	0.481
Kept	Yes	Trigram	0.394
Removed	No	Unigram	0.555
Removed	No	Bigram	0.386
Removed	No	Trigram	0.173
Removed	Yes	Unigram	0.538
Removed	Yes	Bigram	0.380
Removed	Yes	Trigram	0.173

C. WordNet Results

Table III shows the four WordNet configurations. Removing stopwords increased the correlation from about **0.525** to **0.558**. Note that both stopword-removed configurations achieved the same final correlation, and lemmatization had no impact on the final correlation.

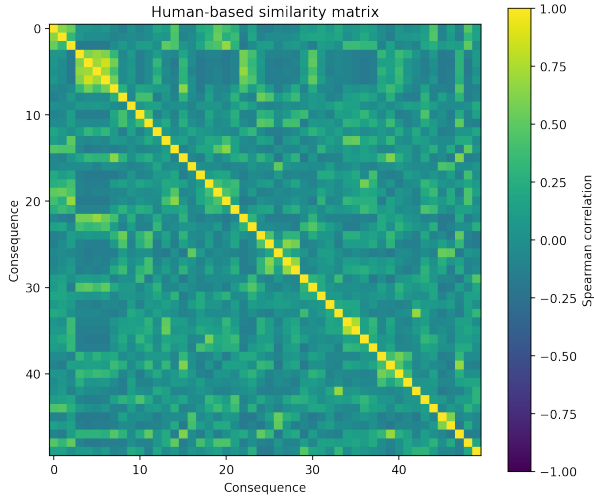
TABLE III
WORDNET CORRELATION WITH THE HUMAN MATRIX

Stopwords	Lemmatization	Pearson r
Kept	No	0.525
Kept	Yes	0.525
Removed	No	0.558
Removed	Yes	0.558

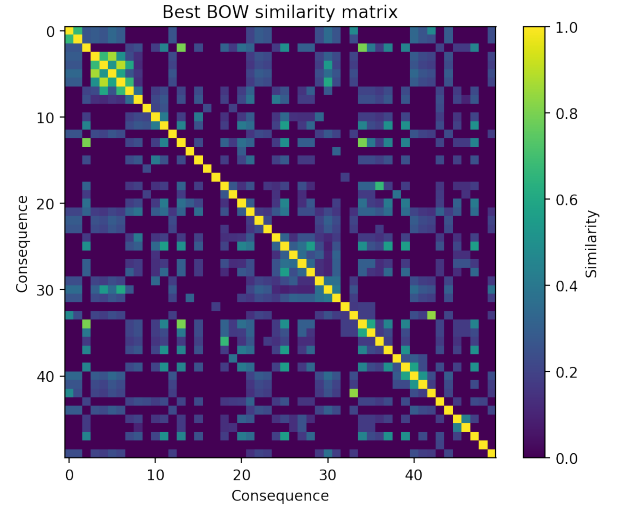
D. Overall Interpretation

The highest score for all 28 NLP matrices was BOW with a score of 0.599. WordNet obtained 0.558, TF-IDF got 0.555. There were not huge differences, but in the results, it can be seen that simple word overlap was very effective for these short description of consequences.

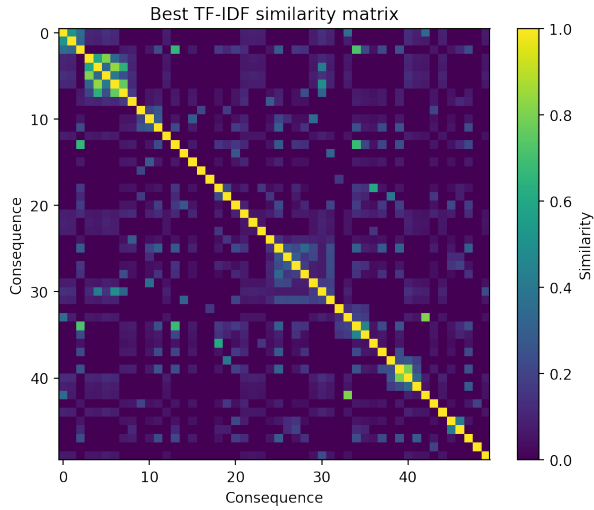
As can be seen in the figures, the distribution of words in BOW and TF-IDF are relatively sparse while the WordNet is denser since related words can also obtain a positive semantic score when they are not exactly the same.



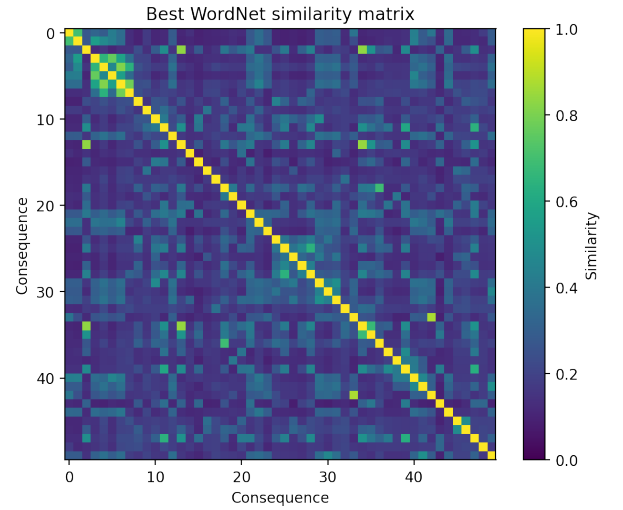
(a) Human-based similarity matrix



(b) Best BOW similarity matrix



(c) Best TF-IDF similarity matrix



(d) Best WordNet similarity matrix

Fig. 1. Human-based similarity matrix and the best BOW, TF-IDF, and WordNet matrices.

The pre-processing results were also beneficial. Stopword removal helped the best unigram configurations for all three techniques, but it caused large decreases for BOW and TF-IDF bigrams and trigrams. This demonstrates the value of trying all the necessary configurations rather than simply hoping that one preprocessing option would always result in a better outcome. One limitation is that the human matrix can contain negative Spearman values, while the NLP similarities used here are nonnegative. Also, the WordNet procedure does not perform word-sense disambiguation and the results must be interpreted in the context of this dataset and should not be used to rank NLP methods in general.

VIII. CONCLUSION

This project aimed to compare human similarity judgment for 50 consequence descriptions of cyberattacks with BOW, TF-IDF, and WordNet similarity. We have developed one human matrix, 24 BOW/TF-IDF matrices, and four WordNet

matrices, and compared all 28 NLP matrices with the human matrix on Pearson's correlation on the upper triangle.

The best BOW result was 0.599 without stop words, no stemming and using a unigram. With the same setting, TF-IDF obtained a value of 0.555, whereas WordNet got 0.558 when stopwords were removed. The results demonstrate the effectiveness of simple unigram word overlap for this dataset, particularly for this grouping.

IX. PROJECT LINK

<https://www.overleaf.com/read/wnzsdhdyjg#6b8116>